PO12Q - Quantitative Political Analysis: Uncovering Relationships

Formulae Collection and Notation

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1 | Formulae

Table 1: Formulae for PO12Q

Statistic	Formula
Correlation Coefficients	
Covariance	$\text{cov}(x,y) = \frac{1}{n-1}\sum(x_i - \bar{x})(y_i - \bar{y})$
Pearson's r	$r_p = \frac{\text{cov}(x,y)}{s_x s_y}$ $= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$
Point-Biserial r_{pb}	$r_{pb} = \frac{\bar{x}_1 - \bar{x}_0}{s}\sqrt{\frac{n_1 n_0}{n(n-1)}}$
Spearman's r_s	$r_s = 1 - \frac{6\sum d_i^2}{n(n^2-1)}$
ϕ Coefficient	$\phi = \frac{AD - BC}{\sqrt{(A+B)(C+D)(A+C)(B+D)}}$

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Statistic	Formula
Correlation – Significance Testing	
Noncentrality Parameter (power)	$\lambda = \frac{\rho\sqrt{n-2}}{\sqrt{1-\rho^2}}$
Pearson (Significance)	$t_{\text{obs}} = \frac{r_p\sqrt{n-2}}{\sqrt{1-r_p^2}}$
Point-Biserial (Significance)	$t = \frac{r_{pb}\sqrt{n-2}}{\sqrt{1-r_{pb}^2}}$
Spearman (Significance)	$t = \frac{r_s\sqrt{n-2}}{\sqrt{1-r_s^2}}$
	$t = \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}$
ϕ (Significance)	$\chi^2 = n\phi^2 \quad (df = 1)$
Regression	
Adjusted R^2	$\bar{R}^2 = 1 - \frac{\sum \hat{\epsilon}_i^2/(n-k-1)}{\sum (y_i-\bar{y})^2/(n-1)}$
Coefficient of Determination	$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} = 1 - \frac{\sum \hat{\epsilon}_i^2}{\sum (y_i-\bar{y})^2}$
Estimated Standard Error of $\hat{\beta}_0$	$\hat{se}(\hat{\beta}_0) = \sqrt{\frac{\sum x_i^2}{n\sum (x_i-\bar{x})^2}}\,\hat{\sigma}$
Estimated Standard Error of $\hat{\beta}_1$	$\hat{se}(\hat{\beta}_1) = \frac{\hat{\sigma}}{\sqrt{\sum (x_i-\bar{x})^2}}$
	$\hat{\text{var}}(\hat{\boldsymbol{\beta}}) = \hat{\sigma}^2 (\boldsymbol{X'X})^{-1}$
Estimated Variance	$\hat{\sigma}^2 = \frac{\sum \hat{\epsilon}_i^2}{n-p}$
Estimator of β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$
Estimator of β_1	$\hat{\beta}_1 = \frac{\sum_{i=1}^{N}(x_i-\bar{x})(y_i-\bar{y})}{\sum_{i=1}^{N}(x_i-\bar{x})^2}$
	$\hat{\boldsymbol{\beta}} = (\mathbf{X'X})^{-1}(\mathbf{X'y})$
Explained Sum of Squares	$\sum (\hat{y}_i - \bar{y})^2$
Residual Sum of Squares	$\sum \hat{\epsilon}_i^2 = \sum (y_i - \hat{y}_i)^2$
Total Sum of Squares	$\sum (y_i - \bar{y})^2$
Variance Inflation Factor	$\text{VIF}_j = \frac{1}{1-\bar{R}_j^2}$

1.1 Matrices

$$\boldsymbol{X'X} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} = \begin{bmatrix} n & \sum x_i \\ \sum x_i & \sum x_i^2 \end{bmatrix}$$

$$(\boldsymbol{X'X})^{-1} = \frac{1}{n\sum x_i^2 - \sum x_i \sum x_i} \begin{bmatrix} \sum x_i^2 & -\sum x_i \\ -\sum x_i & n \end{bmatrix}$$

$$\boldsymbol{X'y} = \begin{bmatrix} 1 & 1 & \dots & 1 \\ x_1 & x_2 & \dots & x_n \end{bmatrix} \times \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = \begin{bmatrix} \sum y_i \\ \sum x_i y_i \end{bmatrix}$$

2 | Notation

Table 2: Notation for PO12Q

Symbol	Explanation
a	A scalar; a single numerical value representing magnitude, written in lower-case italics.
$\boldsymbol{a}$	A vector; an ordered list of numbers (often a column) representing magnitude and direction, written in lower-case bold.
$\boldsymbol{A}$	A matrix; a rectangular array of numbers arranged in rows and columns, written in upper-case bold.
A^{-1}	Inverse of matrix A; the matrix satisfying $AA^{-1} = I$, the matrix analogue of dividing by A.
A′	Transpose of matrix A; the matrix obtained by interchanging its rows and columns.
α	Significance level; the probability of rejecting a true null hypothesis (a Type I error). Common levels are 0.05, 0.01, or 0.001; it also fixes the confidence level $1 - \alpha$.
α_i	Coefficients of an auxiliary (secondary) regression – one regressing an independent variable on the others, used for example to compute the variance inflation factor (VIF) for multicollinearity rather than to answer the main research question.
β	Population regression coefficient; the number multiplied by a variable in the regression equation. A slope coefficient gives the change in y for a one-unit increase in that predictor, holding the others constant.
$\hat{\beta}$	Estimated regression coefficient; the Sample Regression Function estimate of β, obtained by Ordinary Least Squares (OLS), which minimises the sum of squared residuals.
χ^2	Chi-squared test statistic; assesses whether the observed frequencies in a contingency table differ significantly from those expected if there were no association between the variables, $\chi^2 = \sum(O - E)^2 / E$.
d_i^2	Difference in ranks for each pair of observations for Spearman Correlation.
d_{obs}	Observed, standardised effect size (Cohen's d); the magnitude of the difference, expressed in standard-deviation units.
d_{test}	Assumed effect size, for example in a power or sample-size calculation; the value that enters the noncentrality parameter, $\lambda = d_{\text{test}} \cdot \sqrt{n}$.
df	Degrees of freedom, express constraints on our estimation process
E[x]	Expected value of x; the long-run average of a random variable over infinitely many repetitions, i.e. the mean of its probability distribution.
$E[y_i \mid x_i]$	Conditional expectation, reading as 'The expected value of y, given x'.
ϵ	Error term; quantifies the distance between each observation and the corresponding point on the (population) regression line, denoted ϵ_i.
$\hat{\epsilon}$	Residual; an estimate of the error term – the difference between an observation y_i and its estimated value $\hat{y}_i$, denoted $\hat{\epsilon}_i$.
F	F-test statistic; compares explained and unexplained variance to test the joint significance of several coefficients at once.
f^2	Cohen's f^2; the standardised effect size for an F-test, the regression counterpart of Cohen's d. For an overall model $f^2 = \frac{R^2}{1-R^2}$, expressing explained relative to unexplained variance; larger f^2 means a larger effect and thus greater power.
$\frac{\partial f(x)}{\partial x}$	The partial derivative of f of x, with respect to x.
H_0	Null Hypothesis, typically asserting that there is no effect or no difference.
H_1	Alternative Hypothesis, proposes that there is an effect or a difference

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Table 2 – continued

Symbol	Explanation
I	Identity matrix; a square matrix with ones on the diagonal and zeros elsewhere, leaving any matrix unchanged under multiplication.
k	Number of slope coefficients (predictors) in the regression model, excluding the intercept.
K	Total number of regression coefficients including the intercept, so that $K = k + 1$.
λ	Noncentrality parameter; how many standard errors the true effect lies from the null value, $\lambda = d_{\text{test}} \cdot \sqrt{n}$; it centres the non-central t-distribution under the alternative hypothesis.
$\lim_{h \to 0}$	The limit as h approaches zero.
log(x)	Logarithm of the variable x; the exponent to which a base b must be raised to yield x (so $x = \log_b n$ when $b^x = n$). In regression the natural logarithm is typically used, for example for skewed variables.
μ	Population mean of the variable of interest; a parameter – a fixed value characterising the population – estimated by the sample mean $\bar{y}$.
μ_0	Null Hypothesis Value
n	Sample size; the number of observations in the sample.
n_i	Number of cases in each group for Point-Biserial Correlation; n = Total.
Ω	Variance–covariance matrix of the regression errors. Under the standard assumptions $\Omega = \sigma^2 I$, where σ^2 is the mean squared error.
p	p-value, indicates the probability of obtaining a result equal to, or even more extreme than the observed value in the direction of the alternative hypothesis, assuming the null hypothesis is true.
$P()$	Probability of a statement
ϕ	Phi coefficient; measures the association between two binary variables, equivalent to Pearson's r applied to two dichotomies.
r	Correlation coefficient; measures the statistical dependence of two variables by pairwise comparison of values, taking a value between -1 and +1 for strength and direction.
R^2	Coefficient of determination; the proportion of variation in the dependent variable explained by the independent variable(s), $R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}$.
$\bar{R}^2$	Adjusted R^2; a version of R^2 that adjusts for the number of independent variables in a multiple regression, giving a more unbiased estimate of the proportion of variance explained.
r_p	Pearson's product-moment correlation coefficient; a parametric measure of the linear association between two variables (assumes approximate normality).
r_{pb}	Point-biserial correlation coefficient; measures the association between a continuous variable and a binary (dichotomous) variable.
r_s	Spearman's rank-correlation coefficient; a non-parametric measure of the monotonic association between two variables, computed on ranks.
ρ	The hypothesised population correlation under H_1.
s	Sample standard deviation; the square root of the variance, $s = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}}$.
se	Estimated standard error; the standard deviation of the sampling distribution of $\bar{y}$ estimated from the sample, $se = \frac{s}{\sqrt{n}}$ (replacing the unknown σ with s).
$\hat{se}$	Estimated standard error; estimates the standard error of a sample regression coefficient.
σ	Residual Standard Error; measures the average distance that the observed values fall from the regression line, expressed in the same units as the response variable.
σ^2	Error variance; the variance of the regression disturbances ϵ, assumed constant under homoscedasticity, and estimated by the mean squared error of the residuals.

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Table 2 – continued

Symbol	Explanation
t_{obs}	Observed, standardised test statistic that measures how many standard errors a sample statistic (typically a sample mean) lies from a hypothesised population parameter.
u	Numerator degrees of freedom of the F-distribution; the number of slope coefficients being tested, $u = k$.
v	Denominator degrees of freedom of the F-distribution; the residual degrees of freedom, $v = n - k - 1$ (equivalently $n - K$).
VcV	Variance–covariance matrix; a square matrix holding the variances of the estimated coefficients on its diagonal and the covariances between pairs of coefficients in its off-diagonal elements.
$\bar{x}_i$	Mean of the continuous variable for group i for Point-Biserial Correlation.
$\bar{y}$	Sample mean; the sum of the observations divided by their number, $\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i$.
$\hat{y}$	A(n estimated) point on the sample regression line.
y_i	The i-th observation in the sample, where $i = 1, 2, \ldots, n$.